

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular End Semester Examination – Summer 2022

Course: B. Tech.

Branch :EXTC

Semester :IV

Subject Code & Name: BTETPE405B Data Compression & Encryption

Max Marks: 60

Date: 27/08/2022

Duration: 3.45 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q.1 Solve Any Two of the following.

- A) Explain with help of block diagram what Data compression is and Explain different methods of data compression? CO2 6
- B) Write short note on RLE Image and text compression? CO2 6
- C) Apply Shannon-fano coding for following message $[x] = [X_1, X_2, X_3, X_4, X_5, X_6]$ and probabilities are $\{P\} = \{0.30, 0.25, 0.15, 0.12, 0.08, 0.10\}$ also find its efficiency and redundancy. CO2 6

CO2

Q.2 Solve Any Two of the following.

- A) Alphabet with Probabilities $\{0.4, 0.2, 0.2, 0.1, 0.1\}$ for $\{a,b,c,d,e\}$ find the following 1) Huffman codes by keeping the probabilities as high as possible 2)Efficiency 3)Variance CO2 6
- B) For the sequence of alphabets given below demonstrate the encoding process using LZW. Construct the dictionary & obtain the encoded output sequence.
Sequence: w a b b a b w a b b a b w a b b a b w CO2 6
- C) Using Arithmetic Coding Decode the message 0.572 in the given coding model. CO2 6

Symbol	O	S	!
Probability	0.4	0.5	0.1

Q.3 Solve Any Two of the following.

- A) What are the different approaches for compressing an image? Explain Old JPEG Standards? CO2 6
- B) Explain the Differential Pulse code modulation technique. CO2 6
- C) Discuss one dimensional discrete cosine transform? CO2 6

Q.4 Solve Any Two of the following.

- A) Write down MPEG Audio standards. CO2 6

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|---|-----|---|
| B) Define Digital Audio and explain its generation? | CO2 | 6 |
| C) Write a short note on μ - law companding. | CO2 | 6 |

Q. 5 Solve Any Two of the following.

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|--|----------|---|
| A) What are the challenges of computer security? | CO1, CO2 | 6 |
| B) Classify the different type of security attack and explain them with example? | CO1, CO2 | 6 |
| C) Explain the working of DES with the help of block diagram. | CO1, CO2 | 6 |

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